

Q1. Looping statements

CODE

```
#include <stdio.h>
int main()
{
    int num , reverse =0, remainder;
    printf(" Enter a number:");
    scanf("%d", &num);
    while (num ! 0)
    {
        remainder = num % 10;
        reverse = reverse * 10 + remainder;
        num = num / 10;
    }
    printf (" reversed number = %d\n", reverse);
    return 0;
}
```

INPUT 12345

OUTPUT Enter a number: reversed number = 54321

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Q2. Looping statements

CODE

```
#include <stdio.h>
int main()
{
    int n,t,r,rev = 0;
    scanf("%d", &n);
    t =n;
    while (n>0)
    {
        r = n % 10;
        rev = rev*10+r;
```

```

    }
    if (t== rev)
        printf("palindrome");
    else
        printf("Not palindrome");
    return 0;
}

```

INPUT 121

OUTPUT palindrome

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Q3. Looping statements

CODE

```

#include <stdio.h>
int main ()
{
    int num,originalNum, remainder, sum =0;
    printf("Input :");
    scanf("%d", &num);
    originalNum = num;
    while (originalNum ! 0)
    {
        remainder = originalNum% 10;
        sum += remainder * remainder * remainder;
        originalNum/= 10;
    }
    if (sum == num)
        printf ("output: Armstrong\n");
    else
        printf ("output: not Armstrong\n");
    return 0;
}

```

INPUT 143

OUTPUT

Input :output: not Armstrong

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Q4. Looping statements

CODE

```
#include <stdio.h>
int main()
{
    int n,i,j,prime;
    scanf("%d", &n);
    for (i=2; i< n;i+ )
    {
        prime = 1;
        for (j=2;j< i;j+ )
        {
            if (i%j == 0)
            {
                prime = 0;
                break;
            }
        }
        if (prime)
            printf("%d", i);
    }
    return 0;
}
```

INPUT

20

OUTPUT

235711131719

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Q5. Looping statements

CODE

```
#include <stdio.h>
int main()
{
    int n;
    scanf("%d", &n);
    for (int i=1; i< n; i+ )
    {
        for (int j =1; j< n;j+ )
        {
            if(i = 1 | i = n | j = 1 | j = n )
                printf("*");
```

```
            else
                printf("");
        }
    }
    printf("\n");
}
return 0;
}
```

INPUT 5

OUTPUT *****
**
**
**

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Q6. Looping statements

CODE

```
#include <stdio.h>
int main()
{
    int n;
```

```

int n;
scanf("%d", &n);
for (int i = 0; i < n; i+ )
{
    for (int j = 0; j < i; j+ )
        printf ("");
    for ( int j = 0; j < n-i; j+ )
        printf(" *");
    printf("\n");
}
return 0;
}

```

INPUT 4

OUTPUT

```

* * * *
  * * *
  * *
  *

```

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Q7. Looping statements

CODE

```

#include <stdio.h>
int main()
{
    int n;
    scanf("%d", &n);
    for (int i = 1; i < n; i+ )
    {
        for(int j = 1; j < n-i; j+ )
            printf("");
        for (int j =1; j < 2*i - 1; j+ )
            printf("*");
        printf("\n");
    }
}

```

```

    }
    return 0;
}

```

INPUT ₄

OUTPUT *

```

    ***
    *****
    *****

```

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Q8. Looping statements

CODE

```

#include <stdio.h>
int main()
{
    int n;
    scanf("%d", &n);
    for (int i = 1; i < n; i++)
    {
        for (int j = 1; j < n - i; j++) printf(" ");
        for (int j = 1; j < 2 * i - 1; j++) printf("*");
        printf("\n");
    }
    for (int i = n - 1; i > 1; i--)
    {
        for (int j = 1; j < n - i; j++) printf(" ");

```

```

        for (int j = 1; j < 2 * i - 1; j++) printf("*");
        printf("\n");
    }
    return 0;
}

```

INPUT ₃

OUTPUT

Compilation Error:

```
C:\Windows\TEMP\sandbox_192312447RA_6a632a4fc97986.81025276\main.c: In
function 'main':
C:\Windows\TEMP\sandbox_192312447RA_6a632a4fc97986.81025276\main.c:14:34:
error: 'i' undeclared (first use in this function)
   14 | for (int j = 1; j <= n - i; j ++){printf(" ");
      | ^
C:\Windows\TEMP\sandbox_192312447RA_6a632a4fc97986.81025276\main.c:14:34: note:
each undeclared identifier is reported only once for each function it
appears in
```

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Q9. Looping statements

CODE

```
#include <stdio.h>
int main ()
{
    int n, count = 1;
    scanf("%d", &n);
    for (int i = 1; i < n; i++)
    {
        for (int j = 1; j < i; j++)
        {
            printf("%d", count+ );
        }
        printf("\n");
    }
    return 0;
}
```

INPUT ₄

OUTPUT ₁

```
23
456
```

78910

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Q10. Looping statements

CODE

```
#include <stdio.h>
int main()
{
    int n,i,j;
    scanf("%d", &n);
    for (i = 0; i < n; i+ )
    {
        int val = 1;
        for (j = 0; j < i; j+ )
        {
            printf("%d", val);
            val = val * (i - j)/ (j + 1);
        }
        printf("\n");
    }
    return 0;
}
```

INPUT ₄

OUTPUT ₁

11
121
1331

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